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Anatomy accommodating prosthetic inter vertebral disc with lower height

Abstract

An intervertebral disc includes a superior endplate having an upper vertebral contacting surface and a lower bearing surface, wherein the upper vertebral contacting surface of the superior endplate has a central portion that is raised relative to a peripheral portion of the superior endplate, and wherein the lower bearing surface has a concavity disposed opposite the raised central portion. The disc includes an inferior endplate having a lower vertebral contacting surface and an upper surface, wherein the lower vertebral contacting surface of the inferior endplate has a central portion and wherein the upper bearing surface has a concavity disposed opposite the central portion. A core is positioned between the upper and inferior endplates, the core having upper and lower core bearing surfaces configured to mate with the bearing surfaces of the upper and inferior endplates. The upper vertebral contacting surface of the superior end plate has a different shape than the lower vertebral contacting surface of the inferior end plate.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. patent application Ser. No. 17/501,504, filed on Oct. 14, 2021 (published as U.S. Pat. Pub. No. 2022-0031470), which is a continuation of U.S. patent application Ser. No.

16/560,538, filed Sep. 4, 2019, now U.S. Pat. No. 11,730,039, which is a continuation of U.S. patent application Ser. No. 15/230,116, filed Aug. 5, 2016, now U.S. Pat. No. 10,441,431, which is a continuation of U.S. patent application Ser. No. 14/339,280, filed Jul. 23, 2014, now U.S. Pat. No. 9,421,107, which is a continuation of U.S. patent application Ser. No. 13/647,933, filed Oct. 9, 2012, now U.S. Pat. No. 8,808,384, which claims the benefit of U.S. Provisional Application No. 61/546,848, filed Oct. 13, 2011, the entire contents of which are incorporated herein by reference.

BACKGROUND AND SUMMARY OF THE INVENTION

(1) The present invention relates to medical devices and methods. More specifically, the invention relates to intervertebral prosthetic discs and methods of preserving motion upon removal of an intervertebral disc.

(2) Back pain takes an enormous toll on the health and productivity of people around the world. According to the American Academy of Orthopedic Surgeons, approximately 80 percent of Americans will experience back pain at some time in their life. In the year 2000, approximately 26 million visits were made to physicians' offices due to back problems in the United States. On any one day, it is estimated that 5% of the working population in America is disabled by back pain.

(3) One common cause of back pain is injury, degeneration and/or dysfunction of one or more intervertebral discs. Intervertebral discs are the soft tissue structures located between each of the thirty-three vertebral bones that make up the vertebral (spinal) column. Essentially, the discs allow the vertebrae to move relative to one another. The vertebral column and discs are vital anatomical structures, in that they form a central axis that supports the head and torso, allow for movement of the back, and protect the spinal cord, which passes through the vertebrae in proximity to the discs.

(4) Discs often become damaged due to wear and tear or acute injury. For example, discs may bulge (herniate), tear, rupture, degenerate or the like. A bulging disc may press against the spinal cord or a nerve exiting the spinal cord, causing “radicular” pain (pain in one or more extremities caused by impingement of a nerve root). Degeneration or other damage to a disc may cause a loss of “disc height,” meaning that the natural space between two vertebrae decreases. Decreased disc height may cause a disc to bulge, facet loads to increase, two vertebrae to rub together in an unnatural way and/or increased pressure on certain parts of the vertebrae and/or nerve roots, thus causing pain. In general, chronic and acute damage to intervertebral discs is a common source of back related pain and loss of mobility.

(5) When one or more damaged intervertebral disc cause a patient pain and discomfort, surgery is often required. Traditionally, surgical procedures for treating intervertebral discs have involved discectomy (partial or total removal of a disc), with or without interbody fusion of the two vertebrae adjacent to the disc. When the disc is partially or completely removed, it is necessary to replace the excised disc material with natural bone or artificial support structures to prevent direct contact between hard bony surfaces of adjacent vertebrae. Oftentimes, pins, rods, screws, cages and/or the like are inserted between the vertebrae to act as support structures to hold the vertebrae and any graft material in place while the bones permanently fuse together.

(6) A more recent alternative to traditional fusion is total disc replacement or TDR. TDR provides the ability to treat disc related pain without fusion provided by bridging bone, by using a movable, implantable, artificial intervertebral disc (or “disc prosthesis”) between two vertebrae. A number of different artificial intervertebral discs are currently being developed. For example, U.S. Pat. Nos. 7,442,211; 7,531,001 and 7,753,956 and U.S. Patent Application Publication Nos. 2007/0282449; 2009/0234458; 2009/0276051; 2010/0016972 and 2010/0016973 which are hereby incorporated by reference in their entirety, describe artificial intervertebral discs with mobile bearing designs. Other examples of intervertebral disc prostheses are the Charite® disc (provided by DePuy Spine, Inc.) MOBIDISC® (provided by LDR Medical (www.ldrmedical.fr)), the BRYAN Cervical Disc (provided by Medtronic Sofamor Danek, Inc.), the PRODISC® or PRODISC-C® (from Synthes Stratec, Inc.), the PCM disc (provided by Cervitech, Inc.), and the MAVERICK® disc (provided by

Medtronic Sofomor Danek).
(7) These known artificial intervertebral discs generally include superior and inferior endplates which locate against and engage the adjacent vertebral bodies, and a core for providing motion between the plates. The core may be movable or fixed, metallic, ceramic or polymer and generally has at least one convex outer surface which mates with a concave recess on one of the plates in a fixed core device or both of the plates for a movable core device. In order to implant these intervertebral discs, the natural disc is removed and the vertebrae are distracted or forced apart in order to fit the artificial disc in place. Depending on the size of the disc space, many of the known artificial discs have a height which is higher than desired resulting in an unnaturally over distracted condition upon implantation of the disc. For example, a smallest height disc available can be about 10-13 mm for lumbar discs and about 5-6 mm for cervical discs.
(8) Currently available artificial intervertebral discs do not provide a desired low profile for some patients with smaller disc heights. It would be desirable to provide a lower height disc which mimics more closely the natural anatomy for smaller patients.
(9) In addition, the vertebral body contacting surfaces of many of the known artificial discs are flat. The inventor has recognized that this flat configuration does not generally match the surfaces of the vertebral bodies resulting in less than ideal bone to implant contact surfaces. It would be desirable to provide a more anatomically shaped vertebral body contacting surface for an artificial disc.
(10) According to an aspect of the present invention, an intervertebral disc comprises: a superior endplate having an upper vertebra contacting surface and a lower bearing surface, wherein the upper surface of the upper endplate has a domed central portion and wherein the lower bearing surface has a concavity disposed opposite the domed central portion; an inferior endplate having a lower vertebra contacting surface and an upper surface, wherein the lower surface of the lower end plate has a domed central portion and wherein the upper bearing surface has a concavity disposed opposite the domed central portion; a core positioned between the superior and inferior endplates, the core having upper and lower surfaces configured to mate with the bearing surfaces of the superior and inferior endplates; and wherein the domed central portion of the superior endplate has a height greater than a height of the domed central portion of the inferior endplate.
(11) According to another aspect of the present invention, an intervertebral disc comprises: a superior endplate having an upper vertebral contacting surface and a lower bearing surface, wherein the upper vertebral contacting surface of the superior end plate has a central portion that is raised relative to a peripheral portion of the superior end plate, and wherein the lower bearing surface has a concavity disposed opposite the raised central portion; an inferior endplate having a lower vertebral contacting surface and an upper surface, wherein the lower vertebral contacting surface of the inferior endplate has a central portion that is raised relative to a peripheral portion of the inferior endplate and wherein the upper bearing surface has a concavity disposed opposite the central portion; and a core positioned between the upper and inferior endplates, the core having upper and lower core bearing surfaces configured to mate with the bearing surfaces of the upper and inferior endplates; wherein the upper vertebral contacting surface of the superior end plate has a different shape than the lower vertebral contacting surface of the inferior end plate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The features and advantages of the present invention are well understood by reading the following detailed description in conjunction with the drawings in which like numerals indicate similar elements and in which:
(2) FIG. **1**A is a schematic cross-sectional view of a vertebra including an intervertebral disc according to an aspect of the present invention;

(3) FIG. **1**B is a schematic cross-sectional view of a vertebra including an intervertebral disc according to the prior art;
(4) FIGS. **2**A-**2**C are top perspective, side, and side, cross-sectional views, respectively, of an intervertebral disc according to an aspect of the present invention;
(5) FIGS. **3**A-**3**E are top, bottom, side, anterior cross-sectional, and side cross-sectional views, respectively, of a superior endplate of an intervertebral disc according to an aspect of the present invention;
(6) FIGS. **4**A-**4**E are top, bottom, side, anterior cross-sectional, and side cross-sectional views, respectively, of an inferior endplate of an intervertebral disc according to an aspect of the present invention;
(7) FIG. **5** is a side cross-sectional view of an intervertebral disc according to the prior art;
(8) FIGS. **6**A, **6**B, and **6**C illustrate several of various possible geometries for raised central portions of a superior endplate according to aspects of the present invention;
(9) FIG. **7** is a graph of the average radius of curvature of the superior endplate for a collection of patients separated by disc level; and
(10) FIG. **8** is a graph of the average depth of the superior endplate concavity for a collection of patients separated by disc level.

DETAILED DESCRIPTION

(11) FIGS. **2**A-**2**C show an intervertebral disc **27** that comprises superior and inferior endplates **33** and **35** that, as seen in FIG. **1**A, are sized and shaped to fit within an intervertebral space **37**. Each endplate **33** and **35** has a vertebral contacting surface **39** and **41**, respectively, and an inner surface **43** and **45**, respectively. As seen in the cross section of FIG. **2**C, the superior endplate **33** has a first or lower bearing surface **47** on the inner or lower surface **43** of the superior endplate, and the inferior endplate **35** has a second or upper bearing surface **49** on the inner or upper surface **45** of the inferior endplate. The bearing surfaces **47** and **49** are spherical in shape, however other known bearing surface shapes may also be used.
(12) A mobile core **51** is configured to be received between the first and second bearing surfaces **47** and **49**. The core **51** has at least one, ordinarily two, curved bearing surfaces **53** and **55**. The core **51** is ordinarily rigid, i.e., substantially inflexible and incompressible, however, the core may be flexible, compressible, and/or compliant with rigid bearing surfaces. In the embodiment shown in FIG. **2**A-**2**C, the core **51** is rigid and has first and second curved bearing surfaces **53** and **55** for contacting the first and second curved bearing surfaces **47** and **49**, respectively, of the superior and inferior endplates. The superior and inferior endplates **33** and **35** are articulable and rotatable relative to each other via sliding motion of at least one of, ordinarily both of, the first and second bearing surfaces **47** and **49** over the core **51**.
(13) The inventor has recognized that providing raised or domed central portions of both the upper vertebral contacting surface **39** and the lower vertebral contacting surface **41**, the disc can be made smaller and closer in size to the natural disc providing more nearly natural motion. The inventor has also recognized that the raised or domed central portions can be advantageously raised in different amounts, to more accurately match the anatomy of the majority of patients while further reducing the height of the disc. Matching the natural disc height accurately leads to more natural motion while overstuffing a large disc in a small disc space can lead to limited motion.
(14) Discs which are too large for the size of a disc space of a patient are thought to be detrimental to the patient. A connection has been found between fusion (no motion) at a disc level and increased degenerative disc disease at adjacent disc levels. This leads to the conclusion that overly large discs with limited mobility may also result in increases in degenerative disc disease at adjacent levels. Placing a disc which is too large for the disc space can also result in the unwanted result of migration sometimes leading to removal of the disc.
(15) In one embodiment of an intervertebral disc system designed for use in the cervical spine, a core has a diameter of about 8-10 mm, a height of about 4-6 mm, and a spherical radius of

curvature of the bearing surfaces of about 10-18 mm. The intervertebral discs **27** and **29** may be provided in different heights to accommodate patient anatomy. For example, discs may be provided in multiple heights, such as of 4, 5, 6 and 7 mm or in other size variations. The discs may also be provided in different sizes. In one example, for example different lengths in the anterior/posterior direction can be used to accommodate different anatomies. Although different width discs can be used, in the preferred embodiment, the width of the discs is approximately the same for the different disc sizes. The same core **51** and bearing surface configuration is preferably use in the discs of multiple sizes

(16) In another embodiment of an intervertebral disc system designed for use in the lumbar spine, a core has a diameter of about 10-15 mm, a height of about 5-10 mm, and a spherical radius of curvature of the bearing surfaces of about 10-18 mm.

(17) The superior endplate **33** (shown in FIGS. **3**A-**3**E) and the inferior endplate **35** (shown in FIGS. **4**A-**4**E) each have left and right longitudinal sides **331** and **33***r* and **351** and **35***r*, when the disc is viewed from the anterior, and anterior and posterior ends **33***a* and **33***p* and **35***a* and **35***p*.

(18) At least the superior endplate **33** includes a projection **57** which engages with a recess **59** of the core **51** to retain the core between the superior and inferior endplates **33** and **35**. The projection **57** can be in the form of an annular rim having a diameter D**3** (FIG. **3**E) and the recess **59** can be in the form of an annular groove, usually having diameter equal to or slightly less than the diameter D**3**. In the embodiments shown in FIGS. **1-3** and **6**, the projection **57** extends 360° around the core **51**. The core and the projection are ordinarily shaped so that the part of the core below the recess can be passed through an opening defined by the projection in a direction substantially along an axis of the opening defined by the projection. In the embodiments illustrated, the endplate **33** having the retaining ring or projection **57** has been illustrated as the superior endplate. However, it should be understood that the inferior endplate **35** can be provided with a projection instead of or in addition to having the projection on the superior endplate. Ordinarily, however, each endplate will not have a projection to keep thickness of the disc to a minimum and to increase range of motion of the end plate without a projection relative to the core and the endplates relative to each other.

(19) In addition to permitting sliding motion, the core **51** is movable with respect to both of the superior and inferior endplates **33** and **35**. “Movable” is specifically defined for purposes of describing the movability of the core **51** with respect to at least one of the superior and inferior endplates **33** and **35** as meaning that the core is adapted to be displaced in a direction toward or away from at least one of the left and right longitudinal sides **331** and **33***r* and **351** and **35***r* and anterior and posterior ends **33***a* and **33***p* and **35***a* and **35***p* of at least one of superior and inferior endplates. The core **51** is movable with respect to both the superior and inferior endplates when the intervertebral disc system is in use implanted between the vertebrae of a patient. However, the core **51** may have a non-movable position when the disc is being implanted or before completion of a surgical implantation procedure. In a preferred embodiment, the core **51** is movable with respect to both the superior and inferior endplates in both the anterior-posterior direction and the left-right direction.

(20) Although the radius of curvature of the bearing surfaces of the core and the bearing surfaces of the plates **47** and **49**, as shown, are substantially the same (substantially congruent) for purposes of distribution of load and reduced wear, in other embodiments, at least one of the first and second bearing surfaces **47** and **49**, or the bearing surfaces of the core can be modified with a groove, channel, depression or flat on a portion of the bearing surface.

(21) Instead of providing two curved bearing surfaces on the core and corresponding curved bearing surfaces on the superior and inferior end plates, one of the bearing surfaces may be another shape, such as substantially flat (not shown). For example, the top bearing surface on the core below the recess can be flat and the lower bearing surface on the superior endplate above the projection can be flat. The first bearing surface on the superior endplate can be larger than the first bearing surface on the core to facilitate limited translational movement of the core relative to the

superior endplate. If the second bearing surface on the inferior endplate and the second bearing surface on the core are curved, the superior and inferior endplates will be articulable and rotatable relative to each other via sliding motion. In another example, the first bearing surface **47** on the inner surface **43** of the superior endplate and corresponding bearing surface **53** on the core **51** can be flat while the upper bearing surface **49** on the upper surface **45** of the inferior endplate **35** and the corresponding bearing surface **55** of the core can be cylindrically curved in a direction to allow anterior posterior rotation of the upper end plate. Other bearing surface shapes which can also be used depending on the type of motion of the mobile core desired including trough shaped, kidney bean shaped, elliptical, or oval bearing surfaces.

(22) As seen, for example, in FIGS. **2**A, **3**A, **3**C, the superior and inferior endplates **33** and **35** can each include one or more elongated fins or keels **63** and **65**, respectively, on the vertebral contacting surface **39** and **41**, respectively, thereof. If a single, central fin is provided, there will be less cutting of bone required during insertion. However, if multiple, usually two, fins, are provided on one or more of the end plates or if a fin has a break **63**′ between forward and rear components (see FIG. **2**A), there can be more bone contact between the fins and increased fixation, among other possible advantages. The fins **63** and **65** as shown have an angled posterior end for ease of insertion and a plurality of angled slots for improved bone fixation.

(23) The superior and inferior endplates **33** and **35** can be configured to be arranged within a disc space to provide motion in the flexion/extension direction up to a predetermined angle. In one example, the predetermined angle is about ±5 to ±15 degrees, and preferably about ±12 degrees of motion in the flexion/extension direction, i.e., relative angular movement of the anterior and posterior ends **33***a* and **33***p* of the superior end plate with respect to the anterior and posterior ends **35***a* and **35***p* of the inferior end plate. In a presently preferred disc system, for implants placed with the center of the concave bearing surfaces **47**, **49** of the endplates of the two discs **27** and **29** of the disc system about 22 mm apart, the following kinematics can be expected: ±5 degrees of axial rotation and ±12 degrees of flexion/extension. The intervertebral disc system self centers due to the fact that as the core moves away from a neutral centered position, the assembled height of the system gradually increases. The force of the surrounding tissue tries to bring the disc back to the lower height configuration of the neutral position.

(24) As seen in FIGS. **3**B and **4**B, the endplates **33** and **35** can be provided with holes or recesses **91** at an anterior end of the endplates to facilitate removal of the endplates anteriorly in a subsequent surgical procedure if needed.

(25) The upper vertebral contacting surface **39** of the superior end plate **33** (FIGS. **3**A-**3**E) has a central portion **71** that is raised relative to a peripheral portion **73** of the superior endplate. As seen in, e.g., FIGS. **2**B and **3**C-**3**E, the central portion **71** can be raised so that it defines a portion of a sphere, however, as seen in FIGS. **4**A-**4**C, the raised central portion can have generally spherical (FIG. **6**A), stepped (FIG. **6**B), or pyramidal or conical (FIG. **6**C) shape, as well as other shapes that might be determined to be useful. These shapes are described without considering the shape of the keel **63** which will ordinarily also extend upwardly from the raised central portion **71**. The lower bearing surface **47** of the superior endplate **33** ordinarily has a concavity disposed opposite the raised central portion **71**.

(26) The lower vertebral contacting surface **41** of the inferior endplate **35** (FIGS. **4**A-**4**E) has a central portion **75** and the upper bearing surface **49** of the inferior endplate also ordinarily has a concavity disposed opposite the central portion. The central portion **75** of the inferior endplate **35** is ordinarily either not raised or not raised to as great an extent relative to the peripheral portion **77** of the inferior endplate as the central portion **71** of the superior endplate **33** is raised relative to the peripheral portion **73** of the superior endplate. Thus, the shape of the upper vertebral contacting surface **39** of the superior endplate **33** is ordinarily different than the shape of the lower vertebral contacting surface **41** of the inferior endplate **35**. When the central portion **75** of the inferior endplate **35** is raised, like the raised central portion **71** of the superior endplate **33**, it can have

generally spherical (FIG. **6**A), stepped (FIG. **6**B), or pyramidal or conical (FIG. **6**C) shape, as well as other shapes that might be determined to be useful.

(27) The height of the superior end plate **33** measured from an outer peripheral edge **79** of the lower bearing surface **47** of the superior end plate to a top **81** of the raised central portion **71** of the upper vertebral contacting surface **39** is ordinarily greater than a height of the inferior endplate **35** measured from an outer peripheral edge **83** of the upper bearing surface **49** of the inferior endplate to a bottom **85** of the central portion **75** of the lower vertebral contacting surface **41**. The superior endplate **33** height is ordinarily at least 30% greater than the inferior endplate **35** height. The central portion **75** of the inferior endplate **35** is ordinarily at least somewhat raised, and the raised central portion **71** of the superior endplate **33** ordinarily has a height which is about 150% to 300% of a height of the central portion of the inferior endplate.

(28) The lower bearing surface **47** and the upper bearing surface **49** ordinarily each comprise generally circular outer peripheral edges **79** and **83**. The outer peripheral edge **83** of the upper bearing surface **49** can have a larger diameter D**1** (FIG. **4**E) than the diameter D**2** (FIG. **3**E) of the outer peripheral edge **79** of the lower bearing surface **47**.

(29) The upper vertebral contacting surface **39** of the superior endplate **33** and the lower vertebral contacting surface **41** of the inferior endplate **35** may, in addition to the keels **63** and **65**, be provided with knurling, teeth, serrations or some other textured surface **67** to increase friction between the vertebral contacting surfaces and the adjacent vertebra. As seen, for example, in FIGS. **2**A and **3**A, the textured surface **67** can include a plurality of ribs **69** on the upper vertebral contacting surface **39** of the superior end plate **33** that can extend onto the raised central portion **71** of the superior endplate. The raised central portion **71** of the superior endplate **33** is raised relative to what is ordinarily a generally flat, planar peripheral portion **73** of the superior endplate, although the peripheral portion may have a slight slope away from the raised central portion. The teeth and ribs **69** or other structures of the textured surface extend upwardly from the generally planar peripheral portion **73** of the superior endplate **33**. The lower vertebral contacting surface **41** of the inferior end plate **35** may also have a textured surface, such as at least one of teeth and ribs extending downwardly from the generally planar peripheral portion **77** of the inferior endplate.

(30) The upper bearing surface **49** and the lower bearing surface **47** ordinarily each comprise part surfaces of spheres (ordinarily but not necessarily having the same radius R**1**) with generally circular outer peripheral edges **83** and **79**. A core retention portion comprises a wall **61** extending one of downwardly from the outer peripheral edge **79** of the lower bearing surface **47** (shown in FIG. **3**E) and upwardly from the outer peripheral edge **83** of the upper bearing surface **49**.

(31) The upper and lower core bearing surfaces **53** and **55** of the core **51** ordinarily each comprise generally circular outer peripheral edges **87** and **89**, respectively, and the core can have a non-spherical transition region **91** between the upper and lower core bearing surfaces. The recess **59** can be provided in the transition region **91**, such as in the form of an annular groove, and the projection **57** can extend inwardly from the wall **61**, such as in the form of an annular protrusion. A height of the transition region **91** between the upper and lower core bearing surfaces **53** and **55** is greater than a height of the wall **61** of the core retention portion.

(32) By providing the raised central portion **71** on the upper vertebral contacting surface **39** of the superior endplate **33**, the disc **27** can have a reduced height B relative to a disc height C of a conventional disc **127** in which both upper and lower vertebral body contacting surfaces of are flat, as seen by comparing FIGS. **1**A and **1**B and **2**C and **5**. The inventor has recognized that the prior art flat configuration does not generally match the surfaces of the vertebral bodies, which results in less than ideal bone to implant contact surfaces. The superior endplate **33** with the raised central portion **71** provides a more anatomically shaped vertebral body contacting surface for an artificial disc while providing the same endplate to endplate spacing A and an equal range of motion. As seen in FIG. **1**A, by lowering the periphery **73** of the upper vertebral contacting surface **39** of the superior endplate **33** relative to the periphery **173** of a conventional superior endplate **133**, the

upper vertebra **200** is allowed to settle into a lower position relative to the lower vertebra. This can, moreover, be accomplished without changing the clearance (ROM) between end plates or the position of the centers of rotation of the endplates. The raised central portion **71** can, in addition, facilitate fixation because the raised central portion can sit deeper into vertebra behind the external cortex, so it is less likely to expulse or retropulse.

(33) The inventor has also recognized that, when the central portions **71** and **75** of both the upper vertebral contacting surface **39** and the lower vertebral contacting surface **41** are raised, it can be advantageous to have the central portions be raised in different amounts, i.e., domes on both endplates are not equal because the anatomy of the upper and lower surfaces of the vertebrae are of different shapes.

(34) FIG. **7** shows the radius of curvature of the superior end plate of the sampling of patients to be between about 10 mm and 30 mm with an average value of about 20 mm FIG. **7** shows the curvatures of different disc levels with the vertebrae identified by numbers C4-C7 and the vertebral spaces identified by the numbers of the two adjacent vertebrae. FIG. **8** shows corresponding numbers for depths of the superior end plate cavities. As a comparison, the inferior endplates were found to be substantially flat or having such a small curvature that it was generally immeasurable. Based on this data and the interest in designing a smaller height disc, the central portion of the superior endplate **71** has been raised by a greater amount than that of the inferior endplate to most closely accommodate the patient anatomy.

(35) The superior and inferior endplates of the two discs may be constructed from any suitable metal, alloy or combination of metals or alloys, such as but not limited to cobalt chrome alloys, titanium (such as grade **5** titanium), titanium based alloys, tantalum, nickel titanium alloys, stainless steel, and/or the like. They may also be formed of ceramics, biologically compatible polymers including PEEK UHMWPE, PLA or fiber reinforced polymers. The end plates may be formed of a one piece construction or may be formed of more than one piece, such as different materials coupled together.

(36) The core can be made of low friction materials, such as titanium, titanium nitrides, other titanium based alloys, tantalum, nickel titanium alloys, stainless steel, cobalt chrome alloys, ceramics, or biologically compatible polymer materials including PEEK, UHMWPE, PLA or fiber reinforced polymers. High friction coating materials can also be used.

(37) Different materials may be used for different parts of the disc to optimize imaging characteristics. PEEK endplates may also be coated with titanium plasma spray or provided with titanium screens for improved bone integration. Other materials and coatings can also be used such as titanium coated with titanium nitride, aluminum oxide blasting, HA (hydroxylapatite) coating, micro HA coating, and/or bone integration promoting coatings. Any other suitable metals or combinations of metals may be used as well as ceramic or polymer materials, and combinations thereof.

(38) Any suitable technique may be used to couple materials together, such as snap fitting, slip fitting, lamination, interference fitting, use of adhesives, welding and/or the like.

(39) In the present application, the use of terms such as “including” is open-ended and is intended to have the same meaning as terms such as “comprising” and not preclude the presence of other structure, material, or acts. Similarly, though the use of terms such as “can” or “may” is intended to be open-ended and to reflect that structure, material, or acts are not necessary, the failure to use such terms is not intended to reflect that structure, material, or acts are essential. To the extent that structure, material, or acts are presently considered to be essential, they are identified as such.

(40) While this invention has been illustrated and described in accordance with a preferred embodiment, it is recognized that variations and changes may be made therein without departing from the invention as set forth in the claims.

Claims

1. An intervertebral disc comprising: a superior endplate having: an upper vertebra contacting surface including a raised curved area, a lower bearing surface having a concavity disposed opposite the upper vertebra contacting surface, and a first keel disposed on and extending from the raised curved area of the upper vertebra contacting surface, wherein the first keel includes an anterior projection, an angled posterior projection substantially aligned with the anterior projection and a gap that separates the anterior projection from the posterior projection; an inferior endplate having: a lower vertebra contacting surface, and an upper bearing surface having a concavity disposed opposite the lower bearing surface; and a core positioned between the superior and inferior endplates, the core having an upper surface configured to mate with the lower bearing surface, and a lower surface configured to mate with the upper bearing surface.
2. The intervertebral disc of claim 1, wherein the first keel is elongated in an anterior/posterior direction relative to the disc.
3. The intervertebral disc of claim 1, further comprising a second keel disposed on and extending from the lower vertebra contacting surface in a direction outward from the lower vertebra contacting surface.
4. The intervertebral disc of claim 3, wherein the second keel is a single, centrally located keel.
5. The intervertebral disc of claim 3, wherein the second keel is elongated in an anterior/posterior direction relative to the disc.
6. The intervertebral disc of claim 5, wherein the second keel further comprises a gap defining an anterior component of the second keel and a posterior component of the second keel.
7. The intervertebral disc of claim 5, wherein the second keel further comprises an angled posterior end.
8. An intervertebral disc comprising: a superior endplate having: an upper vertebra contacting surface, and a lower bearing surface disposed opposite the upper vertebra contacting surface; an inferior endplate having: a lower vertebra contacting surface including a raised curved area, an upper bearing surface disposed opposite the lower vertebra contacting surface, and a first keel disposed on and extending from the raised curved area of the lower vertebra contacting surface, wherein the first keel includes an anterior projection, an angled posterior projection substantially aligned with the anterior projection and a gap that separates the anterior projection from the posterior projection; and a core positioned between the superior and inferior endplates, the core having an upper surface configured to mate with the lower bearing surface, and a lower surface configured to mate with the upper bearing surface.
9. The intervertebral disc of claim 8, wherein the first keel is elongated in an anterior/posterior direction relative to the disc.
10. The intervertebral disc of claim 8, further comprising a second keel disposed on the upper vertebra contacting surface.
11. The intervertebral disc of claim 10, wherein the second keel is a single, centrally located keel.
12. The intervertebral disc of claim 10, wherein the second keel is elongated in an anterior/posterior direction relative to the disc.
13. The intervertebral disc of claim 12, wherein the second keel further comprises a gap separating an anterior component of the second keel from a posterior component of the second keel.